Data Science 500 Assignment

RICHFIELD
richfield.ac.za

Name and Surname: Njabulo Mafuyeka

Student ITS No: 402421326

Qualification: BSc IT Year of Study: 1st Semester: 2

Assignment due date: 23 October 2026 Date submitted: 21 October 2026

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: N.Mafuyeka Date: 27 September 2026

Richfield Graduate Institute of Technology (Pty) Ltd is registered with the Department of Higher Education & Training as a Private Higher Education Institution under the Higher Education Act, 1997, Registration Certificate No. 2000/HE07/008.

Table of Contents

Introduction 2

Methodology 3

Data collection 3

Data cleaning 3

Data visualisation 3

Predictive modelling 3

Visualisation 5

Distribution of book prices 5

Distribution of star ratings 5

Average book price by genre 5

Book prices by star rating 5

Predictive Modelling 6

Model performance 6

Interpretation of model coefficients 6

Model limits 7

Recommendation 8

Conclusion 9

Reference List 10

Link to Github repository 10

Introduction

This report analyses book data collected from the Books to Scrape website to explore book prices, ratings, availability and genre trends. A dataset of 400 books was collected using web scraping and prepared for analysis by checking for missing values, duplicate records and inconsistent data types.

The analysis uses visualisations to identify patterns in book prices and ratings and to compare average prices across genres. A linear regression model was also developed to estimate book prices using ratings, stock quantities and genres. The goal is to understand the patterns within the dataset, evaluate how well the model predicts prices and identify the limitations of the results.

Methodology

Data collection:

The data was collected from the Books to Scrape website using Python's requests library to retrieve web pages and BeautifulSoup to extract the relevant information. The first 20 catalogue pages were scraped, which produced a dataset of 400 books. The individual book pages were also accessed to collect each book's genre and stock quantity. The collected fields included the book title, price, star rating, availability, genre and stock quantity. The data was stored in a Pandas dataframe.

Data cleaning:

The dataset was checked for missing values, duplicate records and inconsistent data types. No missing values or duplicate rows were found. One book title appeared twice, but the records had different prices and stock quantities, so both of them were kept.

Whitespaces were removed from the relevant text fields and prices were converted from text to numerical values in pounds. Star ratings were converted from words into numerical scores in a range of 1 to 5. The final dataset contained 400 rows and seven columns, including the numerical rating score.

Data visualisation:

Matplotlib was used to create visualisations showing the distribution of book prices, the number of books at each star rating, average prices across the different genres and the price differences between rating groups. Only genres with at least five books were included in the average price comparison to avoid comparing genres represented by only a few records. The graphs were analysed to identify patterns and determine whether book ratings and genres have any relationship with the prices.

Predictive modelling:

A linear regression model was developed to estimate book prices using rating scores, stock quantities and genres. Genre categories were converted into numerical indicator variables, with Fiction used as the reference category.

The dataset was divided into training (80%) and testing (20%) sets. The model was trained on the training data and evaluated on the unseen test data using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-squared. These values were used to analyse prediction errors and determine how well the selected features of the data explained difference in book prices.

Visualisation

Distribution of book prices:

The histogram showed that book prices ranged from around £10 to £60, with no single dominant price point. The highest concentration appeared around £37.50 to £40, while fewer books were within the £30 to £35 interval and the highest price range. This shows a wide spread in book prices within the dataset.

Distribution of star ratings:

All five star ratings were represented in the dataset. One star books were the most frequent, with about 88 books, while four star books were the least frequent, with about 72. The differences across the groups were small, meaning the ratings are broadly and evenly distributed.

Average book price by genre:

Average prices differed across genres. Fantasy had one of the highest average prices at around £40 and Historical Fiction had around £39. Business had the lowest average price at approximately £29 among the genres included in the comparison. Most genres had average prices between £32 and £38. However, these findings are based on the sampled books, instead of the entire catalogue of books.

Book prices by star rating:

The box plot showed similar price ranges across all five star rating groups, with prices consistently being in the range of £10 to £60. The medians and interquartile ranges (IQR) overlapped, which shows that star ratings do not reliably correlate with book price.

Predictive Modelling

The model used rating scores, stock quantities and genres as input features because these variables may be associated with book prices. Genre was converted into numerical indicator variables, with Fiction used as the reference category, to allow the categorical data to be included in the linear regression model.

Model performance:

The linear regression model was analysed using a test set containing 20% of the dataset. It gave a Mean Absolute Error (MAE) of £12.83, a Root Mean Squared Error (RMSE) of £14.89 and an R-squared value of -0.119.

The MAE suggests that the model's predictions differed from the actual prices by about £12.81. The RMSE was higher because it gives greater weight to larger prediction errors. The negative R-squared value shows that predicting the overall mean book price would have been more accurate than using the model.

These results suggest that ratings, stock quantities and genres were not sufficient to predict book prices accurately. This means that the model currently has limited practical value for estimating prices.

Interpretation of model coefficients:

Fiction was used as the reference genre. Politics, Contemporary and Health gave the highest positive coefficients, showing higher predicted prices than Fiction when the other book features were constant. Autobiography and Biography had negative coefficients, showing lower predicted prices relative to Fiction under the same conditions.

Rating Score had a coefficient of -0.12, suggesting a small negative relationship between ratings and predicted prices. Stock Quantity had a coefficient of -0.74, meaning that an increase of one unit in stock quantity was associated with a decrease of about £0.74 in predicted price.

These values describe the associations within the model, instead of proving that ratings, stock quantities or genres cause prices to change. This means that the estimated relationships should be interpreted cautiously, due to the model's poor performance.

Model limits:

The model was trained on 400 books and used only three feature types, which included ratings, stock quantities and genres. Other factors that may influence book prices were not included. The dataset also contained an unusual genre label, “Add a comment”, which may have affected the interpretation of that category.

The negative R-squared value and high prediction errors show that the model requires improvement before it can be used for reliable price estimation. Collecting more data, checking for any unusual category labels in the data and testing alternative models could help improve future results.

Recommendation

The findings suggest that book prices should not be estimated using star ratings, stock quantities and genres alone. The regression model performed worse than just predicting the average price, meaning that the actual prices should be used when accurate pricing decisions are required.

Future analysis could be improved by collecting data from more books and checking unusual category labels before modelling. More book features that are relevant could also be investigated to determine whether they improve prediction accuracy. Other regression models could also be tested and compared using the same values for evaluation.

The results should be interpreted within the sample of the 400 books analysed. A larger sample would provide stronger results for understanding pricing patterns across the full catalogue.

Conclusion

This assignment used web scraping, data cleaning, visualisation and linear regression to analyse 400 books from the Books to Scrape website. The visualisations showed dispersion in book prices and average prices across genres, while the box plot suggested that star ratings were not a strong indicator of price.

The linear regression model produced a Mean Absolute Error of £12.81 and an R-squared value of -0.119, which highlighted a poor performance in its predictions. This means that the selected features were insufficient for reliable price estimation in this dataset.

The analysis of the book data demonstrated how data science techniques can be applied to collect, prepare and investigate data. It also highlighted the importance of data quality, selecting the appropriate data features and evaluating the performance and values produced when drawing conclusions or developing predictive models.

Reference List

- Bro Code (2025a). *Learn Matplotlib in 1 hour!*. [online] YouTube. Available at: https://www.youtube.com/watch?v=c9vhHUGdav0 [Accessed 27 Sept. 2026].
- Bro Code (2025b). *Learn Pandas in 1 hour!*. [online] YouTube. Available at: https://www.youtube.com/watch?v=VXtjG_GzO7Q [Accessed 27 Sept. 2026].
- *Learn NumPy in 1 hour!*. (2025). [Video] Directed by Bro Code. YouTube. Available at: https://www.youtube.com/watch?v=VXU4LSAQDSc [Accessed 27 Sept. 2026].
- *Machine learning in python: Building a linear regression model*. (2020). Directed by Data Professor. Available at: https://www.youtube.com/watch?v=R15LjD8aCzc [Accessed 11 Oct. 2026].
- Ryan & Matt Data Science (2023). *Hands-on linear regression with scikit-learn in Python (beginner friendly)*. [online] YouTube. Available at: https://www.youtube.com/watch?v=ukZn2RJb7TU [Accessed 11 Oct. 2026].
- *Web scraping & automation with python: Extracting book data from "books to scrape"*. (2024). [Video] Directed by kawsarlog. YouTube. Available at: https://www.youtube.com/watch?v=SROnFyxGVJM [Accessed 1 Oct. 2026].
- *Web scraping with python - beautiful soup crash course*. (2020). [YouTube Video] Directed by freeCodeCamp.org. Available at: https://www.youtube.com/watch?v=XVv6mJpFOb0 [Accessed 28 Sept. 2026].

Link to Github repository:

https://github.com/Njabulo-bit-bit/Data-Science-500-Assignment.git